Web CAD 2D Lite

Plumbing Edition

Complete User Guide

Revision 36f

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1. Introduction

Web CAD 2D Lite – Plumbing Edition is a browser-based 2D drafting application for building drawings and MEP installations (plumbing, fire fighting, HVAC, electrical). It runs in the browser without installation, saves your work automatically in the browser, and exports to DXF so it can be opened in AutoCAD and other CAD software.

What this tutorial covers (revision 20)

- All menus: File, Edit, View, Toolbar, Format, Tools, Draw, Dimension, Modify, Insert, Room, Sheet, Objects, MEP, Help.
- Every ribbon tab and the tool panel on the left.
- Every command you can type in the command line (full list in chapter 19).
- New features: Purge, Room (Revit style), Spline, Image (PNG/JPG), Blocks, Sheet Set, Drawing title & scale.

Plans

Plan	Function
Free	All drawing features. Print/PDF output carries a watermark.
Freelancer	Print/PDF without watermark for individuals.
Company	Print/PDF without watermark for teams/companies.

Language

Choose the language with the selector at the top right of the home page, forum, pricing page and workspace: Bahasa Indonesia, English, Bahasa Melayu, Hindi and Chinese. The whole interface (menus, ribbon, tool panel, dialogs, messages) and this tutorial follow your choice. Help > Tutorial opens the PDF in the active language.

> **Note.** Standard values (SNI, NFPA, ASHRAE, PUIL) and automatic generation results are approximations and can be edited. A qualified designer must check all results before they are used for construction.

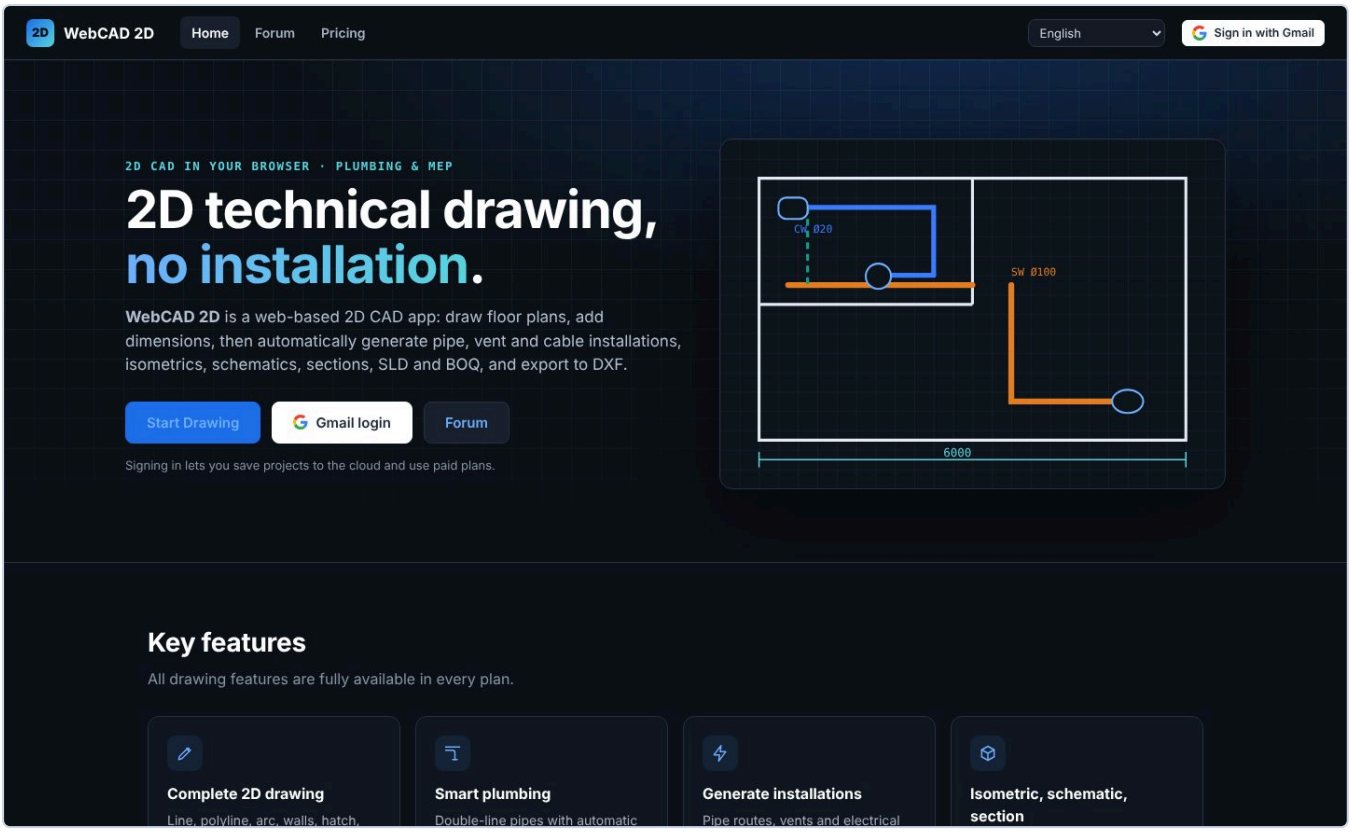


Figure 1. Home page with the language selector.

2. The interface

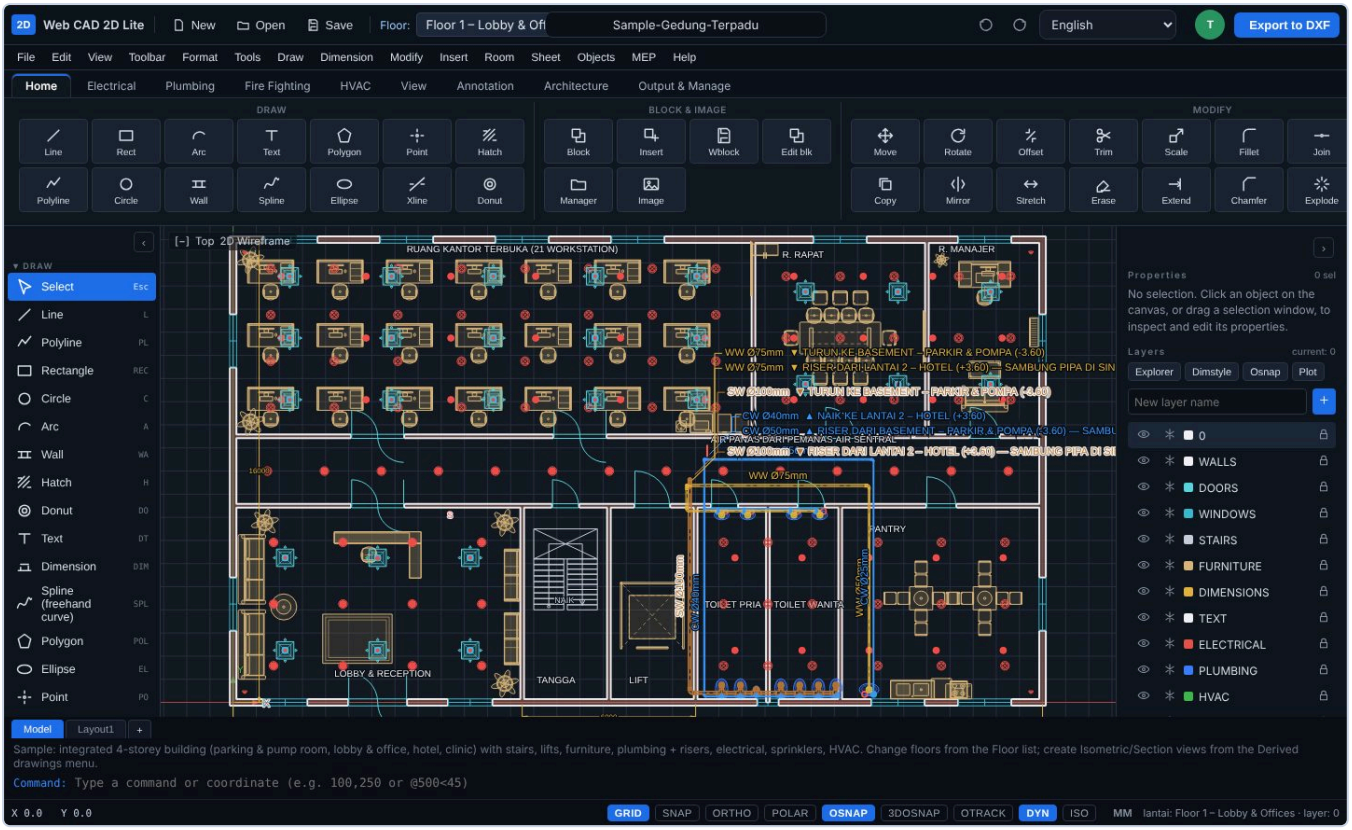


Figure 2. Main interface.

- Menu bar: File, Edit, View, Toolbar, Format, Tools, Draw, Dimension, Modify, Insert, Room, Sheet, Objects, MEP, Help.
- Ribbon: tabs Home, Electrical, Plumbing, Fire Fighting, HVAC, View, Annotate, Architecture, Output & Manage. The contents change with the tab.
- Tool panel (left): tools by group. Click a group title to fold/unfold it; the small arrow minimizes the whole panel. Your choice is remembered.
- Properties (right): properties of the selected object (including Room, blocks, drawing title) and the layer list.
- Command line (bottom): type a command (e.g. LINE, ZE) or coordinates (100,250 or @500<45), then press Enter / Space. Aliases are listed in chapter 19.
- Status bar: GRID, SNAP, ORTHO, OSNAP, DYN, POLAR, OTRACK, ISO, 3D, units, active floor and layer.
- Model / Layout tabs (bottom left): model space and print sheets.

The Objects menu (new)

The Objects menu lists every architectural and MEP object that is also in the ribbon and tool panel: doors, windows, stairs, furniture, pipes, accessories, sanitary fixtures (including pump, roof tank, ground tank, septic tank, infiltration well), fire fighting, HVAC and electrical components. Choose one to activate its tool immediately. The menu is built automatically from the ribbon so it always matches.

Navigation

Action	Function
Pan	Hold the middle mouse button and drag
Zoom	Mouse wheel
Zoom Extents	ZE
Undo / Redo	Ctrl+Z / Ctrl+Y
Cancel command	Esc
Select all	Ctrl+A

Function keys

Key	Function
`F1`	Help / command list
`F2`	Text window
`F3`	Object snap
`F4`	3D osnap (2D adaptation)
`F5`	Isoplane
`F6`	Coordinates: off / absolute / polar
`F7`	Grid
`F8`	Ortho
`F9`	Snap
`F10`	Polar tracking
`F11`	Object snap tracking
`F12`	Dynamic input

3. Drawing basics

Start from the Draw menu, the Home ribbon, the tool panel, or type the command. Every tool shows a prompt in the command line. Left-click picks a point, Enter/Space finishes, Esc cancels. The internal unit is millimetres.

Object	Command
Line	`LINE L`
Polyline	`PLINE PL`
Rectangle	`RECT REC`
Polygon	`POLYGON POL`
Circle	`CIRCLE C`
Arc	`ARC A`
Ellipse	`ELLIPSE EL`
Donut	`DONUT DO`
Point	`POINT PO`
Construction line	`XLINE XL`
Spline (freehand curve)	`SPLINE SPL`
Hatch	`HATCH H`
Text	`TEXT T DT`
Wall	`WALL W`

Closing with C / Close

While drawing LINE, PLINE or WALL, just type C or CLOSE and press Enter: the shape closes to the first point automatically, as in AutoCAD. To finish without closing press Enter or Esc.

Spline

Type SPL (or SPLINE), click the points the curve passes through, type C to close or Enter to finish. SPE (SPLINEDIT) edits the selected spline. The alias SPL was chosen so it does not clash with other commands.

Coordinates and drafting aids

- Absolute coordinates: 100,250. Relative: @500,0. Polar: @500<45.
- Ortho (F8) restricts to straight lines; Polar (F10) guides angles; Object Snap (F3) snaps to endpoint, midpoint, centre, intersection and more; Dynamic Input (F12) shows length and angle next to the cursor.
- Select objects: click, or drag a Window (left to right) / Crossing (right to left). QSELECT selects quickly by type or layer.

Query: distance, area, list, ID

Command	Function
`DI DIST`	Distance and angle between two points
`AA AREA`	Area and perimeter from clicked points
`LI LIST`	Full properties of the selected object
`ID`	Coordinates of a point
`QSELECT`	Quick select by filter
`UN UNITS`	Units, precision, angle

4. Modifying objects

Select objects and run the command, or run the command and then select. The Modify menu and the Modify group of the Home ribbon contain all of them.

Command
MOVE M — move
COPY CO — copy
ROTATE RO — rotate
MIRROR MI — mirror
SCALE SC — scale
OFFSET O — offset
STRETCH S — stretch
TRIM TR — trim
EXTEND EX — extend
BREAK BR — break
FILLET F, CHAMFER CHA — rounded / bevelled corner
JOIN J — join; EXPLODE X — explode
ARRAY AR — rectangular / polar array
LENGTHEN LEN — change length
MATCH MA — match properties
SPE — edit spline; HATCHEDIT — edit hatch
ERASE E / Delete — erase

Note. Double-click an object to edit it directly: text, hatch, Room, drawing title, block (BEDIT).

5. Dimensions, text and drawing titles

The Dimension menu and the Annotate tab contain Aligned (DAL/DIM), Linear (DLI), Radius (DRA), Diameter (DDI) and Angular (DAN) dimensions. Dimension styles are managed in the Dimension Style Manager (D / DIMSTYLE): text height, arrows, precision and scale; choose the active style from the ribbon list.

Text: command TEXT / DT; text styles (font, height, width, oblique, upside-down, vertical) are set in the Text Style Manager (STYLE / DDSTYLE) and exported to DXF. Text on the canvas has a halo so it stays readable over lines and hatches.

Drawing title & scale (TTL / DTITLE)

A tag that names a drawing together with its scale, such as “SECTION A-A SCALE 1:100”. Run TTL, fill in number, title and scale, choose one of 7 designs (Classic, Bubble, Underline, Box, Banner, Arrow, Minimal), then click where to place it. The scale can follow the nearest layout viewport automatically. Double-click the title or run DTITLEEDIT to change it; Properties also shows number, title, scale and design.

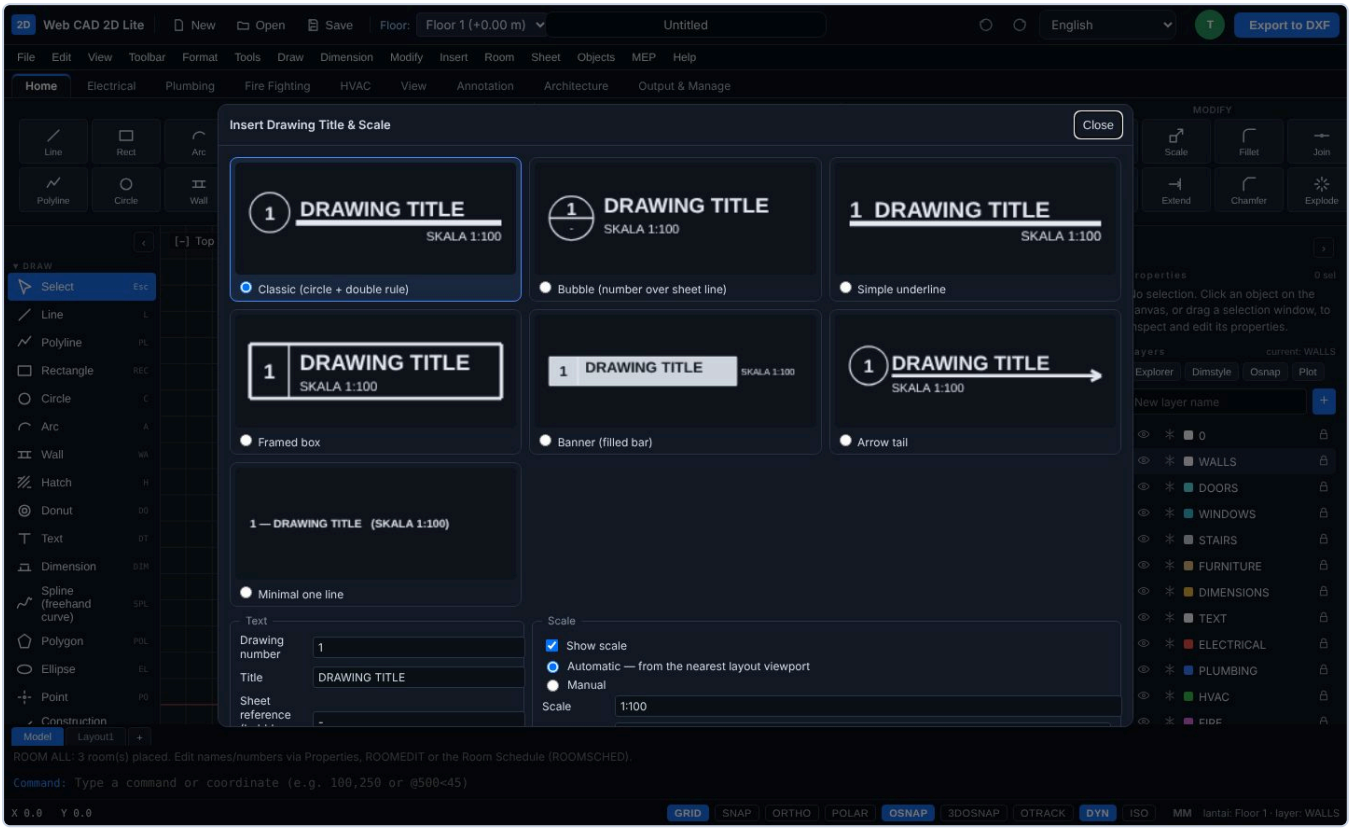


Figure 3. The seven drawing title & scale designs.

6. Layers, linetypes and Purge

The Layer Explorer (LA) shows the columns Name, On, Freeze, Lock, Color, Linetype, Lineweight, Transparency and more. The Layers group in the ribbon provides Make current, Previous, Isolate, Freeze, Lock, Merge, Delete and Purge. The Linetype Manager (LT) manages line patterns; CTB manages plot styles.

Purge (PU / PURGE / -PURGE / PURGEALL)

Purge removes unused items to keep the file light, as in AutoCAD. The dialog lists the categories Blocks, Dimension styles, Layers, Linetypes, Text styles, Zero-length geometry and Empty text; tick any or all, use the Purge nested items option, then Purge Selected or Purge All. -PURGE works from the command line and PURGEALL cleans everything at once. Layer 0, the current layer and items still in use are never removed.

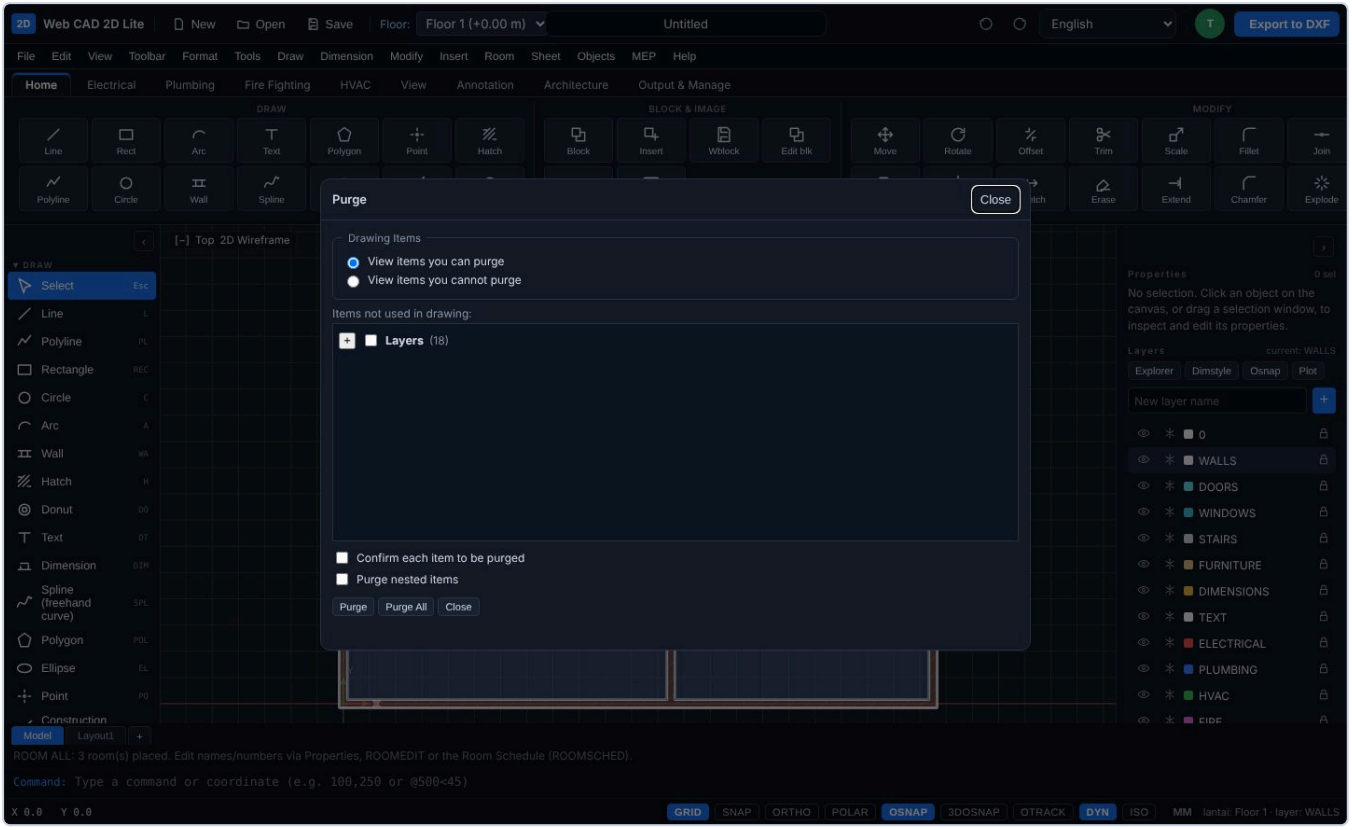


Figure 4. The Purge dialog.

7. Blocks, images and Xrefs

The block features follow AutoCAD. The Insert menu and the “Block & Image” group in the Home ribbon contain all the commands.

Command	Function
`BLOCK B`	Create a block from objects: name, base point, select objects, Retain / Convert / Delete options, Annotative, Scale uniformly, Allow exploding, unit, description.
`INSERT I -INSERT`	Insert a block or DXF file: pick from the list or Browse, set scale, rotation, explode; -INSERT name for the command line.
`WBLOCK W`	Write a block / selected objects to a DXF file (with your own file name).
`BEDIT BE`	Block editor: change the block contents, save, and all insertions update.
`BLOCKS DC`	Block Manager: block list, usage count, rename, delete, export.
`IMAGE IM`	Insert a PNG / JPG image without a border (click to place, drag for size). IMAGEFRAME turns the frame on/off.
`XREF`	Xref Manager: attach another DXF drawing as a reference.

Resizing sanitary and furniture blocks

Select a closet, basin, table, chair, bed or any other furniture, then change Width, Depth or Size (%) in the Properties panel. The drawing, the 3D symbol and the pipe ports follow the new size.

Importing AutoCAD dynamic blocks

Open or insert a DXF file that contains AutoCAD dynamic blocks. The block is imported as the evaluated geometry (the shape as it was saved) and flagged “dynamic”. Dynamic parameters and actions (stretch, flip, visibility state) cannot be recreated; for other shapes, bring the block in as several versions.

8. Floors (levels), copying plans and grids

Like Levels in Revit: every floor has a name and an elevation (mm). Open the Floors dialog with LEVEL / LANTAI, the Tools > Floors menu, or the button in the header. Only Floor 1 exists by default; add floors above or basements below. Objects appear only on the floor where they were drawn; the floor directly below shows faintly as an underlay (reference).

Bringing the floor 1 plan to floor 2 when floor 2 differs

1. Open Floor 1 and select the objects to copy (Ctrl+A for everything).
2. Open the Floors dialog (COPYLEVEL). Under “Copy selected objects to another floor” choose Floor 2 and click Copy.
3. Switch to Floor 2. The copy is independent: change walls, doors or rooms on floor 2 without affecting floor 1.

> **Note.** Risers are not copied. The copy is made once; if floor 1 changes later, floor 2 does not follow. When floor 2 differs only partly, copy everything and edit the differences, or copy in parts (walls first, for example). Use the floor-1 underlay to align positions.

Grid / axes

AXISGRID (AXG) creates a grid from the number of bays and spacing; GRIDEDIT (GE) changes the name, position and spacing of each bay; GRIDLINE (GL) draws a single manual grid line. Grids live on the GRID layer, show on every floor (one shared set for all plans) and export to DXF. To hide the grid, turn off or freeze the GRID layer; this applies to all floors at once.

9. Walls, doors, windows, stairs, furniture and Rooms

Walls (WALL): click points, type C to close. Thickness is set in Properties. Doors (DOOR) and windows (WINDOW) are placed on the wall and cut the opening automatically; change width, type and position in Properties; Flip or Mirror changes the swing. Stairs (STAIR): straight, L, U or spiral with adjustable width, number of treads and rise. Furniture (FURN): a library of 101 items for offices, homes, hotels, hospitals, kitchens, schools/shops and decor; R rotates 90°, M mirrors.

Rooms like Revit

A Room is created from closed walls; doors and windows count as closures. The Room menu and the Room group in the Architecture tab contain all commands. Rooms update automatically when walls move, support Undo/Redo, are saved with the drawing, export to DXF (boundary and text) and appear in the BOQ (section E).

Command	Function
`ROOM RM`	Click inside the walls: the Room is placed. Within the command: A = all, P = polygon, S = separation line, Esc = finish.
`ROOMALL RMA`	Place every enclosed room on this floor; names are taken from texts inside the room.
`ROOMSEP RMS`	Room separation line for open spaces without a wall.
`ROOMPOLY RMP`	Room from a polygon you draw yourself.
`ROOMEDIT RME`	Room properties dialog (or double-click the Room).
`ROOMSCHED RMSCH`	Room schedule: number, name, type, area, perimeter, volume, finishes; CSV export; place as a table on the drawing.
`ROOMTAG RMT`	Show / hide room tags.
`ROOMCOLOR RCS`	Room colour scheme and tag options (name, number, area, type).
`ROOMRENUM RMN`	Renumber rooms sequentially.
`ROOMUPDATE RMU`	Re-detect the boundaries of all rooms.

Room properties in the Properties panel: name, number, type, department, area, perimeter, volume (read-only), height, base offset, floor/wall/ceiling finishes, comments, tag, bounded, fill colour. The Room name and type are used by Generate HVAC (e.g. a room called “Server Room” becomes type server) and by the BOQ.

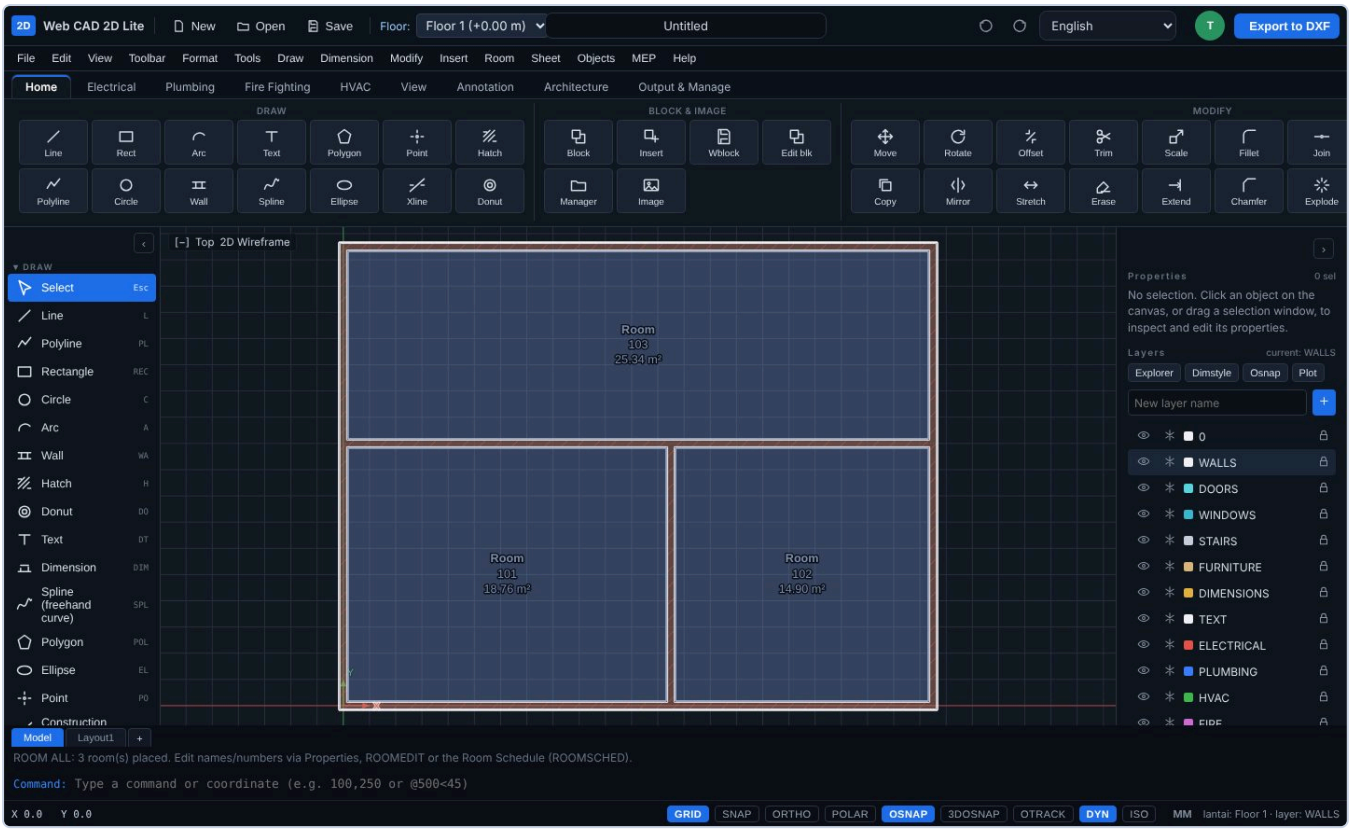


Figure 5. Rooms with tags and a colour scheme.

> **Note.** Limitations: columns inside a room are not subtracted from its area; the Room fill colour is not exported to DXF (boundary and text are); a schedule table placed on the drawing is a static snapshot.

10. Plumbing

The Plumbing tab has the Piping, Pipe Accessories and Sanitary groups. Place the sanitary fixtures first, draw the main lines, then run GENPIPA if you want automatic branch pipes.

Pipes

Tool	Command
Clean water (CW)	`PAB`
Hot water (HW)	`PAP`
Waste water (SW)	`PBE`
Soil water (SP)	`PAK`
Vent	`PV`
Riser	`RISER RS`
Reducer (type the diameter, click mid-pipe)	`RDC`

- Click the start point; the preview follows the cursor and angles snap to 45°. Double line = real diameter with a dash-dot centreline. Pipe sizes in mm or inch via Format > Units (PUNIT).
- Fittings (elbow 90/45 short/long, tee, wye, cross, reducer, socket) are derived from the geometry, one size at each joint. Each segment has its own diameter in Properties; “Apply to the whole run” is available. Soil pipe is at least 3 inch (Ø75).

Sanitary fixtures and equipment

Tool	Command
Sit / squat closet, urinal	`CSD CSJ URN`
Basin, shower, faucet	`WS SH KR`
Floor drain, roof drain	`FD RD`
Pump	`PM`
Roof tank (toren)	`TRN`
Ground tank	`GT`
Septic tank, infiltration well	`ST SR`

- Closets, urinals, basins, showers and faucets can only be placed against a wall (click near the wall). Fixtures cannot overlap: they snap flush. Moving a fixture drags the connected pipes.
- The pipe type must suit the fixture: closet/urinal = clean + soil; basin = clean + hot + waste; shower/faucet = clean + hot; floor/roof drain = waste; pump/roof tank/ground tank = clean; septic = soil; infiltration well = waste. Wrong type → a message appears and the pipe is refused.

Pipe accessories

Gate, ball, butterfly, check valves, PRV, water meter, strainer, backflow preventer, cleanout, air vent, WHA, pressure gauge, union/flange. Place them on a pipe; the alias code shows in the tooltip (e.g. PGV,

PBV, PCK, PRV, WMT).

Generating pipes

GENPIPA creates branch pipes from every fixture port to the nearest main line and sizes the pipes (SNI 03-7065-2005, fixture units / peak flow); the Vent tick box generates vents too. HAPUSGEN removes the generated results on the active floor. The results are ordinary objects that stay editable.

Risers between floors

Riser tool: pick a pipe end, choose up/down and the target floor. A leader and note appear on the source floor; on the target floor the riser appears automatically and follows the parent pipe (position, diameter, type). Waste/soil always go down; vent goes up.

11. Fire Fighting

The Fire Fighting tab has the Fire Fighting group (GENFIRE, sprinkler pipe, hydrant/header pipe, BOQ), Sprinkler & Hydrant (pendant/upright heads, indoor/outdoor hydrant boxes, hose reel, extinguisher, siamese/FDC, control box), and Pump, Tank & Valve (fire tank, diesel+electric+jockey pump set, electric/diesel/jockey pumps, gate/check/alarm check valves, flow switch, pressure gauge).

- GENFIRE places sprinkler heads, indoor hydrants, extinguishers and valves inside the building and routes pipes to the riser point; pumps and tanks are placed manually or with the Pump Set.
- Edit each object's Properties (flow, pressure, hazard class); the BOQ then counts quantities and pipe lengths.

12. HVAC

The HVAC tab has Ducting (GENHVAC, supply/return/fresh-air/exhaust ducts, fittings), Terminals & Equipment (diffusers, grilles, AHU, fans, FCU, louvers, and so on) and Dampers & Controls (volume damper, fire damper, thermostat). GENHVAC calculates the cooling load, supply flow and fresh air (ASHRAE 62.1 / SNI 03-6572) per room, places diffusers, return grilles, fresh-air louvers, exhaust grilles + fans for toilets/kitchens, the AHU, then draws the supply, return, fresh-air and exhaust ducts (with fire dampers at wall penetrations).

> **Tip.** The generator recognises the room type from text or from the Room (office, meeting room, lobby, toilet, kitchen, server, and so on). Draw the walls, place a Room or write the room name with Text, then run GENHVAC. A door in a wall counts as part of the closure.

13. Electrical

The Electrical tab has Electrical installation (GENKABEL, Connect, Check Connections, Schedule, SLD, Wiring) and Electrical components: kWh meter, MCB box, downlights (DL1, DL2), TL, wall lamp (WL), 1/2/3-gang switches and grid switch (S1, S2, S3, SG), 1/2-gang outlets and floor outlet (ST1, ST2, STF), and generator (GEN). Switches, outlets and wall lamps must sit on a wall.

- GENKABEL creates lamp-switch groups, circuits, NYM cable routes and MCBs from the lamp/outlet points (PUIL 2011 / SNI 0225:2011; ampacity 1.5–35 mm²). Connect: kWh → MCB → switch → lamp manually (BIM).
- JADWAL: panel schedule (circuit, points, VA, current, MCB, cable, length, voltage drop) with CSV export. SLD: single line diagram. WIRING: panel wiring diagram. Both read the circuit data and stay in sync.

14. BOQ, MEP checks and synchronisation

- BOQ (MEP menu): a table of No, Item Name, Quantity, Unit, Note, grouped as A. Sanitary fixtures, pipes, accessories, electrical, fire, HVAC, up to E. Rooms & finishes (Room). Pipe length is net with no cutting waste.
- MEP check & summary (CKMEP): a list of errors and warnings (unconnected pipes, fixtures without pipes, wrong sizes, unconnected electrical objects); CEKKONEKSI marks electrical objects that are not connected. Check overlapping sanitary and Snap sanitary together are in the Tools menu.
- All sanitary, electrical and MEP objects are BIM objects: the Properties panel shows category, name, mounting height and technical data that you can edit.

15. Derived views, VG and visual styles

- ISO (isometric), SKEMATIK (riser diagram) and POTONGAN / SEC (section: pick two cut-line points) are generated automatically from the model and appear in the Floors list (group “Derived drawings”). The isometric viewpoint is chosen with ISOMODE (SW, SE, NE, NW) plus front/right/back/left elevations.
- Two-way realtime: changing the plan updates the derived views almost instantly; moving/deleting pipes, fixtures, risers or walls in the isometric or section changes the plan. SINKRON / SYNCVIEWS refreshes everything. MEP objects cannot be drawn in a derived view; edit the data on the plan.
- VG (Visibility / Graphics): show or hide each system; quick buttons Plumbing only, Fire only, HVAC only, Electrical only and Show all. Visual styles: 2D Wireframe, Conceptual, Hidden, Realistic, Wireframe (View menu) affect how walls look.

16. Layouts, Mview, plotting and Sheet Sets

- The Layout tabs (bottom left) create print sheets. MVIEW creates viewports; each viewport has its own view source (floor plan, isometric, schematic, section), scale and visible systems. Double-click a viewport to move focus into it.
- PLOT: Page setup, printer/plotter (PDF), paper size, orientation, plot area (Display/Extents/Layout/Window), offset, scale, plot style (CTB), watermark on the Free plan. PAGESETUP manages page setups. PUBLISH makes a single multi-page PDF from Model and Layouts.

Sheet Set Manager (SSM) like AutoCAD

1. Open it with SSM or menu Sheet > Sheet Set Manager. Create a sheet set with NSS (New Sheet Set wizard): choose “Existing drawings” (the layouts of this drawing) or “An example sheet set”, fill in name, location and details, choose layouts, confirm, Finish.
2. The panel has the Sheet List, Sheet Views and Model Views tabs. Right-click the sheet set / subset / sheet for New Subset, New Sheet, Import Layout as Sheet, Rename & Renumber, Move Up/Down, Remove, Properties, Publish (PDF / printer), Insert Sheet List Table and Close Sheet Set. Double-click a sheet to open it.
3. A sheet set can be exported and opened as a .wcss file (OPENSHEETSET). Importing an AutoCAD .dst file is best-effort: the proprietary database cannot be read exactly, so the app finds the sheet names and you confirm them in the wizard.

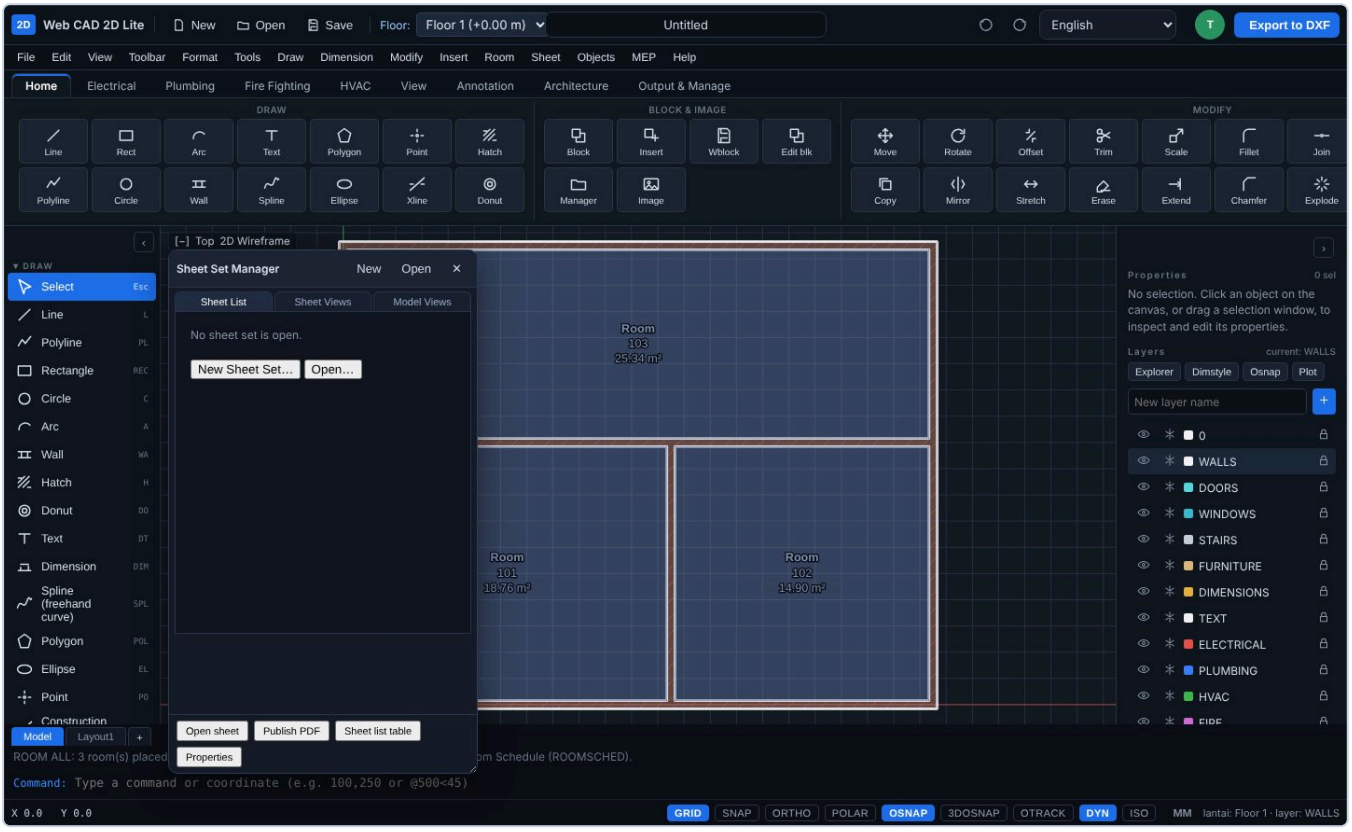


Figure 6. The Sheet Set Manager.

17. DXF export and import

- File > Save (DXF) / Ctrl+S and the DXF/Export button: choose AutoCAD 2007 (AC1021, default), 2018+ (AC1032), 2000 (AC1015) or R12 (AC1009), then Plan / All views. File > Open DXF reads ASCII DXF from AutoCAD 2007 to 2026 (legacy code pages and UTF-8; TEXT/MTEXT).
- BIM objects, blocks, Rooms (boundary and text), text and dimensions are exported. Images and the Room fill colour are not exported. Binary DXF is not read yet (save as ASCII DXF); DWG is not supported directly in the browser, use DXF.

18. Account, plans and forum

- Sign in with Gmail or email. The profile icon in the header shows your name, plan, remaining quota, save to cloud, cloud projects, upgrade and sign out. Drawings are saved automatically in the browser and, when the cloud is on, synchronised to your account.
- Community forum (Help > Forum): discussion list, categories, search, sort, replies, quotes, likes, edit/delete your own posts, and image/DXF/PDF attachments. Plans: Free (DXF export after an ad), Freelancer and Company; payment through Mayar.

19. Command list

Type a command in the command line and press Enter. Some commands have short aliases (shown side by side). The list below matches the help window (F1) and covers the commands in every menu, ribbon and tool panel.

Command	Function
`GENFIRE`	Generate Fire Fighting: sprinkler heads (SNI 03-3989 / NFPA 13), hydrants (SNI 03-1745 / NFPA 14), siamese, control box, fire tank, diesel + electric + jockey pump set (NFPA 20, NFPA 22), pipes and pipe sizes
`GENHVAC`	Generate HVAC: cooling load, fresh air (ASHRAE 62.1 / SNI 03-6572), diffusers, return grilles, supply / return / fresh / exhaust ducts, AHU and duct sizes
`PSP / PHY`	Sprinkler pipe / hydrant pipe (header)
`SPP / SPU`	Sprinkler head pendant / upright
`HYI / HYO / FDC / VBX`	Indoor hydrant box / outdoor hydrant box / Siamese connection / control box with gate valve
`FT / PMS / PME / PMD / PMJ`	Fire tank / pump set / electric pump / diesel pump / jockey pump
`GV / CV / ACV / FS / PG`	Gate valve / check valve / alarm check valve / flow switch / pressure gauge (installed on the pipe body)
`DSA / DRT / DFA / DEA`	Duct supply air / return air / fresh air / exhaust (type 500×300 = width x height)
`DFS / DFR / DFL / RG / EG / LOU / EF / AHU`	Supply diffuser 4-way / round / linear / return grille / exhaust grille / fresh air louver / exhaust fan / AHU
`VCD / FDM`	Volume damper / fire damper (installed on the duct body)
`PAB / PAP / PBE / PAK / PV`	Clean / hot / waste / soil / vent pipes (double line, Ø = actual diameter)
`RDC`	Reducer: click in the middle of the pipe, type a number = end-side diameter
`RS / RISER`	Riser: click a free pipe end, choose up/down and the target floor

Command	Function
`LEVEL / LV`	Manage floors (name, elevation, underlay, copy to floor)
`MEPCHECK`	Check MEP system consistency and summarize pipes, fittings and cables
`SANCHECK / SANFIX`	Check overlapping fixtures / auto-slide them along the wall until flush
`BOQ / GENBOQ`	Bill of Quantity: number, item name, quantity, note (table + CSV + table in the drawing)
`GENVENT`	Generate vent: same as Generate pipes with the Vent option on
`SLD / WIRING`	Draw the panel Single Line Diagram and Wiring Diagram from circuit data
`ISOSW / ISOSE / ISONE / ISONW`	Quick isometric from the SW / SE / NE / NW angle (creates an isometric drawing if none exists)
`DENAH / PLAN`	Back to the plan from a derived drawing
`VPORTS [1|2V|2H|3R|3L|3T|3B|4|J]`	Split the workspace into several tiled viewports (like AutoCAD); each viewport can choose its own view. J = merge
`HUBUNG / CONNECT`	Connect kWh → MCB → switch → lamp / socket by clicking; the result is stored in the BIM data and cables are regenerated
`CEKKONEKSI`	Check and mark electrical objects that are not connected
`TAMPAKDEPAN / KANAN / BELAKANG / KIRI`	Elevations of pipes & MEP from outside the building (FRONT, RIGHT, BACK, LEFT)
`ISOMODE`	Choose an isometric viewpoint (Top/Bottom x Left/Right Front/Rear) like AutoCAD/Revit
`FURN / FU`	Full furniture library (office, home, hotel, hospital, kitchen, etc.); R = rotate 90°, M = mirror
`SYNCVIEWS`	Refresh all derived drawings (isometric, schematic, section) from the plan; plan and derived drawings update each other in real time

Command	Function
`GENKABEL`	Generate electrical installation: lamp-switch groups, circuits, NYM cable routes and MCBs (PUIL 2011 / SNI 0225:2011)
`GENPIPA`	Generate sanitary branch pipes to the mainline + pipe sizes (SNI 03-7065-2005)
`JADWAL`	Panel schedule: circuits, load, current, MCB, cables, voltage drop
`ISO / SKEMATIK`	Automatic derived drawings: pipe isometric and schematic (riser diagram)
`POTONGAN / SEC`	Pipe section: draw a cut line on the plan and the section is created automatically
`VIEWS / SINKRON`	Manage derived drawings / sync all with the model
`SEPTIC / SR`	Septic tank (soil water) / soakaway well (waste water)
`LINE / L`	Line segments
`PLINE / PL`	Polyline, C closes
`RECTANG / REC`	Rectangle
`POLYGON / POL`	Poligon beraturan: jumlah sisi, titik pusat atau E (Edge), lalu I (di dalam lingkaran) / C (di luar lingkaran) dan radius
`CIRCLE / C`	Circle
`ARC / A`	3-point arc
`ELLIPSE / EL`	Elips: dua titik ujung sumbu lalu jarak sumbu lain, atau C untuk mulai from titik pusat
`DONUT / DO`	Donut
`POINT / PO`	Point
`XLINE / RAY`	Construction line
`TEXT / DT / MTEXT`	Text
`HATCH / H`	Hatch dialog: angle, scale, pick point or select
`GRADIENT`	Gradient fill (inside Hatch)
`PUBLISH`	Plot several layouts to one PDF

Command	Function
`VSCURRENT`	Visual style: 2D Wireframe, Conceptual, Hidden, Realistic, Wireframe
`WIREFRAME`	Visual style Wireframe
`HIDE`	Visual style Hidden
`CONCEPTUAL`	Visual style Conceptual
`REALISTIC`	Visual style Realistic
`2DWIREFRAME`	Visual style 2D Wireframe
`PAGESETUP`	Page setup manager
`XREF`	Attach a DXF or image reference
`IMAGE IAT IM IMAGEFRAME`	Attach a PNG/JPG image (no border; Ctrl+V paste or dragging a file onto the canvas also works) / show-hide border
`WALL / WA`	Double-line wall
`DOOR / WINDOW`	Opening on a wall
`DIMALIGNED / DIM`	Aligned dimension
`DIMLINEAR / DLI`	Horizontal or vertical dim
`DIMRADIUS / DRA`	Radius dimension
`DIMDIAMETER / DDI`	Diameter dimension
`VPVIEW / VV`	Choose the view shown in a layout viewport
`DIMANGULAR / DAN`	Angular dimension, 3 points
`SELECTALL`	Select every visible object (Ctrl+A)
`ALIAS / CUI`	Edit command shortcuts
`DIMSTYLE / D`	Dimension style manager
`MOVE COPY ROTATE`	M CO RO
`MIRROR SCALE STRETCH`	MI SC S
`OFFSET TRIM EXTEND`	O TR EX
`BREAK FILLET CHAMFER`	BR F CHA
`JOIN EXPLODE ARRAY`	J X AR
`SETBYLAYER`	Set the selected objects' color, linetype and lineweight to ByLayer

Command	Function
`ARRAY / AR / ARRAYCLASSIC`	Seperti AutoCAD: pilih objek, Enter, lalu R (Rectangular: baris, kolom, jarak antar baris dan kolom atau unit cell dengan dua klik) atau P (Polar: titik pusat, jumlah item, sudut isi, putar objek Y/N)
`FCO / CCO / AUTOCO`	Floor Clean Out (tertanam di lantai) / Ceiling Clean Out (di atas plafon) di ujung pipa air bekas / kotor; AUTOCO memasangnya otomatis pada semua ujung bebas di lantai aktif
`LEGEND / LEGENDUPD`	Add a legend per system (Plumbing, Fire, HVAC, Electrical): symbol, name, Qty, unit; zero Qty is hidden. LEGENDUPD refreshes the counts
`CLASH / BENTROK`	3D clash check between pipe routes (clean, hot, waste, soil, vent, hydrant, sprinkler): list, location, route / elevation change suggestions
`SIMULASI / SIM`	Animated MEP system simulation with timer: water arrival time at each fixture, flow velocity, over-long routes, valves closed/open, gate valve suggestions
`SAMPLEMEP / CONTOHMEP`	2-floor MEP sample plan applying all features (vent, sloping, FCO/CCO, hydrant, sprinkler, HVAC, legend, simulation, clash check)
`SNI / ACUAN`	SNI reference list (plumbing, hydrant, sprinkler, HVAC, PUIL, septic) and the working values used by the app
`TEE / FLIPFLOW`	TEE: sambung vent/cabang ke pipa lain dengan fitting Tee/TY otomatis. FLIPFLOW: balik arah aliran pipa terpilih tanpa memutar pipa
`AUTODIM / ADIM`	Dimensi otomatis denah: rantai bukaan, as dinding, dan total luar pada 4 sisi; dijalankan ulang akan mengganti dimensi otomatis sebelumnya
`DCO / DBA`	Dimensi Continue (rantai from dimensi linear terpilih) dan Baseline (bertumpuk from titik awal sama). Enter mengakhiri

Command	Function
`NORTH / SCALEBAR / ELEVMARK`	Panah utara, bar skala (mengikuti SKALA), dan tanda elevasi sesuai elevasi lantai aktif
`REVCLOUD / AWAN`	Revision cloud from polyline/rect tertutup terpilih atau klik titik lalu Enter
`SKALA / ANNOSCALE [n]`	Skala anotasi 1:n: tinggi teks 2.5 mm, panah, ekstensi, dan stempel dihitung from skala cetak
`DWTAG / DWSCHED`	Tag pintu/jendela otomatis (P1, J1, BV1) dan tabel jadwal pintu & jendela yang diperbarui otomatis
`LAYGEN`	Generator Layout: isi daftar gambar (judul, lantai / gambar turunan, layer, skala, kertas) lalu tekan Buat. Layout, kop, viewport, nomor gambar, cover, dan daftar isi terbentuk otomatis; area besar pada skala tetap dibagi ke beberapa lembar
`LM / KOP / COVER / TOC`	Manajemen layout: LM = manajer lembar (nomor, judul, disiplin, revisi, urutan); KOP = kop gambar 4 template dengan logo dan bagian yang dapat diatur; COVER = cover 3 gaya; TOC = daftar isi gambar otomatis; LAYSYNC = perbarui semua
`MLEADER (MLD) / WIPEOUT (WO)`	MLEADER: klik ujung panah, klik titik shelf, ketik teks (tinggi 2,5 mm kertas sesuai SKALA). WIPEOUT: area penutup warna kertas from titik-titik atau polyline / rectangle tertutup terpilih
`RECTWALL (RW) / BOUNDARY (BO)`	RW: 2 klik sudut membuat 4 dinding. BO: polyline tertutup from area di sekeliling titik klik, menampilkan luas
`NOTASI / PIPELABEL`	Set pipe label notation per system: type (SW/WW/CW/HW), size, material, sloping, BOP, centerline elevation, length, flow arrows
`PIPESPEC / PSPEC`	Spesifikasi pipa: material bawaan tiap sistem (PVC AW / PVC D, PPR PN-10 / 16 / 20 / 25, HDPE, GIP, tembaga, dll)
`DDPTYPE / PTYPE / PDMODE`	Point Style: pilih 20 gaya titik dan ukurannya (relatif layar atau satuan gambar)

Command	Function
`ATTDEF ATT ATTEDIT ATE`	Atribut blok seperti AutoCAD: dialog Attribute Definition (Mode, Tag, Prompt, Default, titik sisip, justifikasi, tinggi teks); jadikan satu blok bersama objek lain, INSERT menanyakan nilai tiap atribut, ATTEDIT / klik ganda untuk mengubah
`BLOCK INSERT WBLOCK BEDIT BSAVE BCLOSE BLOCKS`	AutoCAD-style blocks: Block Definition dialog (base point, objects, behavior, unit), Insert (scale/rotation/explode/Browse DXF), Write Block, Block Editor, Block Manager (rename/delete/export)
`LENGTHEN MATCHPROP`	LEN MA
`ERASE / E`	Erase
`ZOOM / Z`	Zoom ala AutoCAD: ketik opsi A / E (extents), P (previous), W (window), atau skala seperti 2 dan 0.5X; roda mouse = real time
`ZE / ZA`	Zoom extents langsung
`ZOOMW / ZW`	Zoom window
`PAN / P`	Pan
`REGEN / REDRAW`	Redraw
`LAYER / LA`	Layer explorer
`LAYOFF LAYON`	Off layer / all on
`LAYFRZ LAYTHW`	Freeze / thaw
`LAYLCK LAYULK`	Lock / unlock
`LAYISO LAYUNISO`	Isolate / restore
`LAYMCUR / LAYERP`	Make the selected object's layer current / return to the previous layer state
`LAYNEW LAYNEWVP LAYDEL LAYMRG LAYPURGE`	New layer / new (frozen in new viewports) / delete the object's layer / merge into the current layer / purge empty layers
`DSETTINGS / OS`	Snap, grid, osnap
`PLOT / PRINT`	Plot / PDF
`OPEN`	Open DXF or JSON
`QSAVE / SAVE`	Save to cloud (Firebase)

Command	Function
`SAVEAS`	Save as new cloud project
`DXFOUT / EXPORTDXF`	Export DXF
`MODEL / MSPACE / PSPACE`	Model and paper
`MVIEW / MV`	Make viewport
`TILEMODE / LAYOUT`	Switch model / layout
`DIST ID LIST AREA`	Inquiry
`BLOCK / B`	Save selection as a block, then Enter the name
`INSERT / I`	Insert a saved block
`LAYOUT NEW DEL REN`	Add, delete, or rename a layout
`DWG`	AutoCAD DWG cannot be written here — DXF is saved instead
`QSELECT`	Select by type
`SPLINE SPL SPE`	Spline through points (click points, C = close, U = remove last point, Enter = finish) / SPLINEDIT converts a polyline into a spline
`C / CLOSE`	During LINE / PLINE / WALL / SPLINE: type C then Enter to close back to the start point
`ROOM RM`	Revit-style room: click inside walls (A = all, P = polygon, S = separation line, Esc = finish). Double-click a room to edit
`ROOMALL RMA`	Place every enclosed room on this level (names are taken from texts inside)
`ROOMSEP RMS`	Room separation line (closes open spaces without a wall)
`ROOMSCHED RMSCH`	Room schedule: number, name, type, area, perimeter, volume, finishes; CSV export, place as table
`ROOMEDIT RME ROOMTAG RMT`	Room properties dialog / show-hide tags
`ROOMCOLOR RCS ROOMRENUM RMN ROOMUPDATE RMU`	Color scheme & tag options / renumber / re-detect bounfromes
`TTL DTITLE`	Drawing title & scale tag (7 designs, automatic scale from viewport)

Command	Function
`SSM NSS OPENSHEETSET SHEETLIST`	Sheet Set Manager / new sheet set wizard / open .wcss (.dst best-effort) / sheet list table
`B I W BE BLOCKS`	Block dialog / Insert dialog / Write block / Block editor / Block manager
`IM IMAGEFRAME`	Attach PNG / JPG image (no border) / toggle image border
`PURGE -PURGE PURGEALL`	Purge dialog (blocks, dimstyles, layers, linetypes, text styles, zero-length geometry, empty text) / command-line version / purge all
`UNITS COLOR`	Units and ByLayer color
`UNDO REDO`	U / REDO, also Ctrl+Z / Y
`STYLE / DDSTYLE / TS`	Text Style Manager: font, tinggi, lebar, miring, terbalik, vertikal
`UNITS / UN / DDUNITS`	Drawing Units: tipe & presisi panjang/sudut, arah dasar, insertion scale, UCS icon
`AXISGRID / AXG / SUMBU / GR`	Create grid / axes automatically (count & spacing per span, Revit style)
`GRIDEDIT / GE`	Edit grid: nama, posisi, jarak antar bentang
`GRIDLINE / GL`	Draw a single manual grid line with a named bubble
`POLAR [sudut] / OTRACK / ISOSNAP / ISOPLANE / 3DOSNAP`	Polar tracking, object snap tracking, snap isometrik (F5), 3D object snap
`TEXTSCR / UCSICON / COORDS / TEXTHALO`	Text window (F2), ikon UCS, mode koordinat (F6), halo teks
`NEW SAMPLE`	New drawing / sample plan

20. Troubleshooting and FAQ

Problem	Function
Blank page when opening from inside the zip.	Extract the zip first, then open index.html.
Room: message “Not enclosed”.	The walls do not close tightly. Close the gap (type C while drawing walls), place a door/window in the door gap, or add a ROOMSEP separation line.
Generate HVAC: “draw the walls first”.	Draw closed walls and name the rooms (Text or Room), then run again.
A fixture will not place.	Closets, urinals, basins, showers and faucets must be on a wall; click closer to the wall.
A pipe is refused.	The pipe type does not match the fixture it connects to (see chapter 10).
An object is not visible.	Check the active floor, layer (On/Freeze) and VG; run SHOWALL and ZE.
The file feels heavy.	Run PURGEALL to remove unused items, then save.
A button or menu seems “missing”.	The Toolbar menu shows/hides the Ribbon, Tools panel and Properties; every tool is also in the Objects menu and the tool panel.

Short FAQ

- Is the grid the same on every floor? Yes, one grid set is shared by all floors (chapter 8). Hiding it = turning off the GRID layer.
- Is my drawing saved? Automatically in the browser (localStorage); export DXF or save to the cloud for a backup.
- Can it open DWG? Not directly; save from AutoCAD as DXF (ASCII) and open that.
- Internal unit? Millimetres; the display units are set in UN / Units.
- Contact and support: Help > Contact & support.

> **Note.** Known limitations: .dst import is best-effort; AutoCAD dynamic blocks import as evaluated geometry; images are not exported to DXF; results have not been validated in real AutoCAD; technical standards (SNI/NFPA/ASHRAE/PUIL) are approximations.

21. How to use each command (AutoCAD style)

This chapter explains what each command does and the steps to use it. It works like AutoCAD: type the command (or its alias) in the command line and follow the prompt line. Type the option letter shown in square brackets, e.g. [Rectangular/Polar], then Enter. Enter or right click = finish / accept the default <...>. Esc = cancel. Points can be clicked or typed as x,y (e.g. 1000,500) or relative to the last point: @dx,dy (e.g. @300,0) or @distance<angle (e.g. @500<45).

Drawing

Command (alias)	Function
`LINE (L)`	Lines. Click the start point, then the next points; type C to close back to the start, Enter / Esc to finish.
`PLINE (PL)`	Polyline. Point by point; C closes, U removes the last point, Enter finishes. A closed polyline can be hatched and offset.
`RECTANG (REC)`	Rectangle: click the first corner, then the opposite corner (or type x,y).
`CIRCLE (C)`	Circle: click the centre, then click or type the radius. Type D to enter a diameter.
`ARC (A)`	3-point arc: start point, second point (on the arc), end point.
`ELLIPSE (EL)`	Ellipse as in AutoCAD: click the two endpoints of one axis, then click or type the distance to the other axis. Type C first to start from the centre (centre, axis endpoint, other-axis distance).
`POLYGON (POL)`	Regular polygon: type the number of sides + Enter; click the centre (or type E to define one edge with two points); choose I (Inscribed) or C (Circumscribed), then click or type the radius.
`DONUT (DO)`	Donut: inside diameter, outside diameter, then click the placement points.
`POINT (PO)`	Point: click its position. The look is set in DDPTYPE (Format > Point style): 20 styles (dot, none, plus, cross, tick, plus circle and/or square) and the size relative to the screen or in drawing units.
`XLINE (XL)`	Infinite construction line: two points.
`SPLINE (SPL)`	Smooth curve through points: click the points, C closes, U removes the last point, Enter finishes.
`HATCH (H)`	Hatch. Choose pattern, angle and scale in the dialog. Pick internal point mode: click inside a closed area. The area is found from lines, arcs, polylines, rectangles, circles and walls - even several separate lines, as long as they form a closed loop. If there is a gap, a message says no closed boundary was found.
`TEXT (DT / T)`	Text: click the insertion point, type the text, Enter.
`WALL (WA)`	Double-line wall: click points, type C to close. Doors (DOOR) and windows (WINDOW): select a wall, then click the position.

Modifying objects

Command (alias)	Function
`MOVE (M)`	Select objects, Enter; base point; second point (or type a distance with ortho on).
`COPY (CO / CP)`	Like MOVE but the original stays; click destination points repeatedly, Enter to finish.
`ROTATE (RO)`	Select objects, Enter; base point; click or type the angle. Options: C = rotate a copy, R = Reference (reference angle then new angle).
`MIRROR (MI)`	Select objects, Enter; two points of the mirror line; answer Y (erase source) or N (keep source, default).
`SCALE (SC)`	Select objects, Enter; base point; type the scale factor, or R (Reference): reference length then new length.
`STRETCH (S)`	Select the objects with a crossing window (right to left), Enter; base point; second point.
`OFFSET (O)`	Type a distance then Enter (or T = Through, distance follows the cursor), click the object, click the side. Works on lines, walls, pipes, circles, arcs, rectangles, open / closed polylines, polygons and ellipses. ARRAY results can be offset too because they are ordinary objects.
`TRIM (TR)`	Click the part of the line / wall / pipe to remove; the cutting edges are all objects that cross it.
`EXTEND (EX)`	Click the boundary line, then click the end of the line to extend.
`BREAK (BR)`	Click the object (the click is the first break point), then click the second break point: the part in between is removed. Type F to choose the first point again.
`FILLET (F)`	Type R then the radius (0 = sharp corner), T for Trim / No trim mode, then click the two lines.
`CHAMFER (CHA)`	Type D then the first and second distances, then click the two lines.
`JOIN (J)`	Select several connected lines, Enter: they are joined into one.
`EXPLODE (X)`	Select objects, Enter: polylines, rectangles, blocks etc. are broken into their basic parts.
`LENGTHEN (LEN)`	Click near the end of a line, then type the length to add (negative = shorten).
`ARRAY (AR)`	As in AutoCAD (ARRAYCLASSIC): select objects, Enter, then choose the type. R = Rectangular: number of rows, columns, distance between rows, distance between columns (negative = left / down) or two clicks as a unit cell. P = Polar: centre point, number of items (including the original), angle to fill (360 = full circle, negative = clockwise), rotate objects Y / N.
`MATCHPROP (MA)`	Click the source object, then the destination object: layer, colour, linetype are copied.
`ERASE (E / DEL)`	Select objects then Enter, or click objects one by one.

View and drafting aids

Command (alias)	Function
`ZOOM (Z)`	Type Z then an option: A or E (show the whole drawing), P (previous view), W (window: two clicks), or a scale such as 2 or 0.5X. Mouse wheel = real-time zoom, hold wheel = pan. ZE = zoom extents straight away.
`PAN (P)`	Drag to pan the view.
`F3 / F8 / F9 / F10 / F11`	Object snap, Ortho, Snap, Polar tracking, Object snap tracking. Full settings in DSETTINGS (OS).
`LAYER (LA), COLOR (COL), LINETYPE (LT), LTSCALE (LTS), LWEIGHT (LW)`	Layer manager, current colour, linetype, linetype scale, lineweight.
`STYLE (ST), UNITS (UN), DIMSTYLE (D)`	Text style, drawing units, dimension style.
`DIST (DI), ID, LIST (LI), AREA (AA)`	Inquiry commands: distance, coordinates, object properties, area.

Note. Aliases that changed from the old habits: ST is now STYLE (text style) as in AutoCAD, and LT = LINETYPE. The septic tank uses SEPTIC, downlight lamps use DL1 / LIGHT. Aliases can be changed at any time with ALIAS (CUI). If you saved your own aliases, ALIAS keeps using yours.

22. Pipes: specification, slope, fittings and revision 25 updates

Pipe material per system

Open Pipe specification (PIPESPEC or PSPEC; also in the MEP menu). For each system - clean, hot, waste, soil, vent, fire, hydrant - choose the default material: PVC AW, PVC D, PPR PN-10 / PN-16 / PN-20 / PN-25, HDPE PE100, GIP, copper, stainless, cast iron, PP, PEX, steel ASTM A53 Sch 40 / 80. Tick Show the material in the pipe label to show it on the plan (e.g. WW Ø 75mm PVC AW i=1%). Tick Apply to existing pipes to replace old materials. Each pipe can still be changed in the Properties panel.

Elevation and slope

- Select a pipe; in Properties fill Pipe elevation (mm) and Slope (%). The fall direction can be toward the end or the start point. Start / end elevations are calculated automatically.
- The button Apply slope to the whole run gives the same slope to every connected pipe of the same system; branches fall toward the outlet.
- On the plan the pipe label shows system, size, material and slope (i=...%) with an arrow for the fall direction. In sections and isometric views the pipe is drawn sloped according to its elevations.

Moving the pipe label

Select a pipe: besides the end and middle grips there is an orange diamond grip. Drag it to move the label along and away from the pipe. The position is saved with the pipe.

Automatic fittings: elbow, tee, reducer

- Pipe sizes follow the standard DN list (15, 20, 25, 32, 40, 50, 65, 75, 80, 90, 100, 125, 150, 200, ...). 45° and 90° elbows are always a single size.
- If one pipe next to a fitting is made smaller, a reducer forms on that side and the fitting keeps the large size. If all pipes meeting at a fitting are changed, the fitting shrinks too and the old reducers are merged back.
- A tee whose branch is smaller than the run becomes a reducing tee (e.g. 4×2) without an extra reducer. With a third pipe of another size, reducers are made on the left and right of the fitting.
- Straight pipes drawn by hand with the same size, system and material are merged into one whole pipe, like generated pipes.
- Isometric: elbows are drawn curved with sockets, tees and crosses have a cylindrical body, reducers are cones. Section: pipe ends joined to another pipe get a socket collar.

> **Note.** Pipe sizes in the app are nominal sizes (DN). PVC / PPR pipes have an outside diameter (OD) that differs from DN; check the manufacturer table when preparing the BOQ.

Symbols and walls

- kWh meters, MCB panels, hydrant boxes, hose reels and fire extinguishers sit flush on the wall face (the back of the box is exactly on the surface, not inside the wall thickness). The text stays readable (not upside down).

- Floor drains and roof drains placed or moved near a wall are pushed out automatically until they no longer touch the wall thickness.

22a. Revision 36: general MEP, realistic 3D, working drawings

General MEP toolbar

The new General MEP tab holds features that apply to every system: Riser & Connections, System Drawings (Isometric, Schematic, Section), Analysis & Checks (BOQ, clash), MEP Documents and System Display. The Fire Fighting and Piping tabs now contain only system-specific tools. The Sanitary group is split into Sanitary, Pumps & Tanks and Wastewater Treatment.

Tee, flip flow and waste rules

Command (alias)	Function
`TEE / TEEJOIN / SAMBUNG`	Click near where two pipes meet (e.g. a vent onto a waste pipe). A Tee/TY fitting is created automatically and the pipe is split at the junction; a pipe end with a small gap is joined to the host pipe.
`FLIPFLOW / FLIP / ALIRAN`	Reverse the flow direction of the selected pipe without rotating it. Elevation and slope are mirrored; run it again to restore.

> **Note.** Waste pipes may join the soil pipe before the septic tank (TY fitting). A sanitary branch from a floor drain / basin straight to the soil main is still rejected.

More realistic 3D viewer

- Curved elbows with the same short/long radius as the isometric view, couplings, conical reducers and tees/TY with sockets. They are included in IFC and OBJ export.
- New 3D models: gate valve, check valve, alarm valve, pressure gauge, flow switch, fire damper and VCD damper.
- Smoother sanitary fixtures: closet, basin and urinal profiles are smoothly interpolated; the sitting closet is now a skirted bowl (full body down to the floor).

Stairs, doors and windows

- New stairs: Half-landing stair, Winder stair and Ramp (slope in %).
- New doors: glass, 2-way, fire, pocket, bifold, rolling, garage, revolving. New windows: casement, 3-panel, louver, awning, vent.
- New properties: Sill height (mm) and Opening height (mm); shown correctly in plan, 3D and export.
- Furn > structure: square / round / steel / KP columns, beam, footing, glass / steel railing, parapet. Column height follows the level height.

Working Drawing (Architecture tab)

Command (alias)	Function
`AUTODIM / ADIM`	Automatic plan dimensions on 4 sides: opening chain, wall axes and overall outer size. Running it again replaces the earlier automatic dimensions.

Command (alias)	Function
`DCO / DBA`	Continue and Baseline dimensions from the selected linear dimension. Enter ends.
`SKALA / ANNOSCALE [n]`	Set the annotation scale 1:n. 2.5 mm paper text height, arrows, extensions and stamps are computed automatically.
`NORTH, SCALEBAR, ELEVMARK`	North arrow, scale bar labelled SKALA 1 : n and an elevation mark using the active level elevation.
`REVCLOUD / AWAN`	Revision cloud from a selected closed polyline, or click points then Enter.
`MLEADER (MLD) / WIPEOUT (WO)`	MLEADER: click the arrow tip, click the shelf point, then type the text (2.5 mm paper height per SKALA). WIPEOUT: a paper-coloured masking area from clicked points or a selected closed polyline / rectangle.
`DWTAG / DWSCHED`	Automatic door / window tags (P1, J1, BV1) and a schedule table that updates itself when doors are added or removed.
`RECTWALL (RW) / BOUNDARY (BO)`	RW: two corner clicks create 4 walls. BO: closed polyline from the area around the click, area is reported.

Note. Layout plotting is now scale-exact: a 1:100 viewport prints 6000 mm as 60 mm on paper (previously a margin made the scale slightly off). The sheet-set title block follows ISO 5457 practice: 20 mm binding margin, 0.7 mm frame, DRAWN / CHECKED boxes.

22b. Layout Pro: layout management, title block, cover, contents

New Layout tab

The Layout tab in the ribbon has four groups: Layout Management, Title Block, Cover & Contents, and Viewport & Output. All sheets, title blocks, the cover and the contents are managed from one place.

Layout Manager (LM)

Type LM or click Manager. The table lists each layout with name, drawing number, title, discipline, revision, date, plus Incl. (listed in the contents) and TB (attach title block) checkboxes. ▲ ▼ reorder, ⧉ duplicates, ⌖ opens the layout, ✕ deletes. The New layout row adds a sheet and attaches the title block at once. Renumber assigns numbers per discipline (A-01, M-01, P-01 and so on).

Title block (KOP)

- Templates: Bottom strip (full), Right panel (vertical), Bottom-right box and Slim strip. A live preview shows in the dialog.
- Parts: each part can be shown or hidden (party block, logo, project, title, drawing number, scale, date, revision, signatories, extra fields, revision table, notes) to suit each company.
- Parties & Logo: add or remove parties (MEP Consultant, MEP Contractor, Owner and others), fill in role, company name, address / phone / email and upload a PNG / JPG logo.
- Apply to the active layout or all drawing layouts. The viewport is fitted to the drawing area outside the title block. The title block profile is saved with the project and becomes the default for the next one.

Command (alias)	Function
`LM / LAYOUTMGR`	Layout manager.
`KOP / TITLEBLOCK`	Title block dialog (template, parts, parties & logo, project, revisions).
`KOPPIHAK / KOPREV`	Jump to the Parties & Logo / Revisions tab.
`KOPLEPAS`	Detach the active layout's title block for manual editing.
`COVER`	Drawing cover: 3 styles, party logos, project data, document type, paper.
`TOC / DAFTARISI`	Automatic contents / drawing list, grouped by discipline.
`LAYSYNC`	Refresh title blocks, cover and contents now.

Automatic cover and contents

COVER creates a cover layout (classic, modern or formal) with the document type, project name, location, every party's logo and a party table. TOC builds a drawing list with number, title, scale, paper size, revision and date. Both are linked to the Layout Manager: adding, renaming, reordering or deleting layouts, or changing numbers and revisions, updates the contents immediately. When rows exceed one page, extra contents pages are added automatically. The cover and contents plot as paper sheets only (no model) through PLOT and PUBLISH.

Layout Generator (LAYGEN)

To avoid creating viewports one by one, type LAYGEN or click Generator. Fill in the drawing list in order: title, discipline, source (floor or derived drawing), scale (Auto or fixed), paper and visible layers. Quick buttons: + All floors, + All derived drawings, + From current screen, + From selected objects (select one zone in the model first). Press Create: every row becomes a complete layout with title block, a viewport on the chosen source and layers, the scale and an automatic drawing number; the cover and contents are created or updated too. At a fixed scale, an area larger than the sheet is split automatically into several sheets marked (1/n). Undo reverts the whole result.

Title block images and MVIEW snapping

Logos of all parties, signatures (Drawn, Checked, Approved) and stamps can be uploaded as PNG or JPG in the Title block dialog (Parties & Logo and Project tabs); the images are saved with the project and reused on every layout. When you create a viewport with MVIEW, drag its corner or move it, the corners snap (OSNAP) to title block lines and corners, paper objects and other viewports. Make sure OSNAP is on.

Worksheet tabs (several drawings at once)

Above the drawing area is a row of tabs, one per worksheet. Click + (or the New button, command NEWTAB) for a new worksheet, or the Open button to open a DXF/JSON file in a new tab. Click a tab to switch; each tab keeps its own Undo history, view and layouts. A yellow dot on a tab means unsaved changes. Click × (or middle-click) to close: if unsaved you get Save & Close, Close without saving, or Cancel. Right-click a tab for Close, Close others, Close all, Duplicate tab and Rename; drag a tab to reorder. Projects from the Cloud Projects menu open in a new tab. All tabs are still there after the page reloads.

Command (alias)	Function
`LAYGEN / GENLAYOUT`	Automatic layout generator from a drawing list.

> **Tip.** Give each layout a number and discipline in the Layout Manager, then click Renumber so the contents are tidy per discipline.

23. Reference: all menus

Below is the complete content of every menu in the menu bar with its command and shortcut. The Objects menu is built automatically from the ribbon so it always matches the ribbon and the tool panel.

File

Menu / item	Command	Shortcut
New	`NEW`	`Ctrl+N`
Open DXF...	`OPEN`	`Ctrl+O`
Save to Cloud	`QSAVE`	`Ctrl+S`
Save as new cloud project...	`SAVEAS`	
Export DXF...	`DXFOUT`	
DWG... AutoCAD uses DXF	`DWG`	
Plot...	`PLOT`	`Ctrl+P`
Publish...	`PUBLISH`	
Page Setup Manager...	`PAGESETUP`	
Xref Manager...	`XREF`	
Sample plan	`SAMPLE`	
Acuan SNI	`SNI`	
Sample MEP (semua fitur, SNI)	`SAMPLEMEP`	
Sheet Set Manager	`SSM`	
Purge...	`PU`	

Edit

Menu / item	Command	Shortcut
Undo	`U`	`Ctrl+Z`
Redo	`REDO`	`Ctrl+Y`
Erase	`E`	
Select all	`SELECTALL`	`Ctrl+A`
Copy	`CO`	
Match properties	`MA`	

View

Menu / item	Command	Shortcut
Visual style: 2D Wireframe	`2DWIREFRAME`	
Visual style: Conceptual	`CONCEPTUAL`	

Menu / item	Command	Shortcut
Visual style: Hidden	`HIDE`	
Visual style: Realistic	`REALISTIC`	
Visual style: Wireframe	`WIREFRAME`	
Zoom extents	`Z`	
Zoom window	`ZW`	
Pan	`P`	
Regen	`RE`	
Model	`MODEL`	
Layout1	`LAYOUT`	
New layout	`LAYOUTNEW`	
Rename layout	`LAYOUTREN`	
Delete layout	`LAYOUTDEL`	
Model in viewport	`MS`	
Paper space	`PS`	

Toolbar

Menu / item	Command	Shortcut
✓ Ribbon		
✓ Tampak		
✓ MEP		
✓ Draw		
✓ Modify		
✓ Annotate		
✓ Hatch		
✓ Arch		
✓ Layers		
✓ Output		
✓ Tools panel (left)		
✓ Properties panel (right)		

Format

Menu / item	Command	Shortcut
Layer explorer...	`LA`	
Color (ByLayer)	`COLOR`	

Menu / item	Command	Shortcut
Linetype...	`LINETYPE`	
Units...	`UN`	
Point style...	`DDPTYPE`	
Text style...	`STYLE`	
Grid / Axis...	`AXISGRID`	
Dimension style...	`D`	

Tools

Menu / item	Command	Shortcut
Drafting settings...	`OS`	
Text window	`TEXTSCR`	`F2`
Distance	`DI`	
ID point	`ID`	
List	`LI`	
Area	`AA`	
Quick select...	`QSELECT`	
Floors (Levels)...	`LEVEL`	
MEP check & pipe summary (BOQ)...	`MEPCHECK`	
Cek sanitary bertumpuk	`SANCHECK`	
Auto-pack sanitary fixtures	`SANFIX`	
Command aliases...	`ALIAS`	
Purge	`PU`	
Make block...	`B`	
Insert block...	`I`	
Xref...	`XREF`	
Appearance & colors...	`THEME`	
Reset to default settings...	`RESETSET`	

Draw

Menu / item	Command	Shortcut
Line	`L`	
Polyline	`PL`	
Rectangle	`REC`	
Polygon	`POL`	

Menu / item	Command	Shortcut
Circle	`C`	
Arc	`A`	
Ellipse	`EL`	
Donut	`DO`	
Point	`PO`	
Construction line	`XL`	
Text	`DT`	
Hatch	`H`	
Wall	`WA`	
Room (after walls)	`ROOM`	`RM`
Spline	`SPL`	`SPL`
Image (PNG / JPG)...	`IM`	`IM`
Drawing title & scale...	`TTL`	`TTL`
Grid / Axis...	`AXISGRID`	
Grid line	`GL`	

Dimension

Menu / item	Command	Shortcut
Aligned	`DIMALIGNED`	
Linear	`DLI`	
Radius	`DRA`	
Diameter	`DDI`	
Angular	`DAN`	
Dimension style...	`D`	

Modify

Menu / item	Command	Shortcut
Move	`M`	
Copy	`CO`	
Rotate	`RO`	
Mirror	`MI`	
Scale	`SC`	
Stretch	`S`	
Offset	`O`	

Menu / item	Command	Shortcut
Trim	`TR`	
Extend	`EX`	
Break	`BR`	
Fillet	`F`	
Chamfer	`CHA`	
Join	`J`	
Explode	`X`	
Array	`AR`	
Lengthen	`LEN`	
Edit spline	`SPE`	
Erase	`E`	

Insert

Menu / item	Command	Shortcut
Block... (make from objects)	`B`	`B`
Insert block...	`I`	`I`
Write block (DXF / library)...	`W`	`W`
Define attributes... (ATTDEF)	`ATTDEF`	`ATT`
Edit block attributes... (ATTEDIT)	`ATTEDIT`	`ATE`
Block editor	`BE`	`BE`
Block manager...	`BLOCKS`	`DC`
Image (PNG / JPG)...	`IM`	`IM`
Image border on/off	`IMAGEFRAME`	
Xref Manager...	`XREF`	
Drawing title & scale...	`TTL`	`TTL`

Room

Menu / item	Command	Shortcut
Room (click inside walls)	`ROOM`	`RM`
Place all rooms	`ROOMALL`	`RMA`
Room separation line	`ROOMSEP`	`RMS`
Room by polygon	`ROOMPOLY`	`RMP`
Room properties...	`ROOMEDIT`	`RME`
Room schedule...	`ROOMSCHED`	`RMSCH`

Menu / item	Command	Shortcut
Show / hide room tags	`ROOMTAG`	`RMT`
Color scheme & tag options...	`ROOMCOLOR`	`RCS`
Renumber rooms...	`ROOMRENUM`	`RMN`
Update rooms to walls	`ROOMUPDATE`	`RMU`

Sheet

Menu / item	Command	Shortcut
Sheet Set Manager	`SSM`	`SSM`
New sheet set...	`NSS`	`NSS`
Open sheet set (.wcss / .dst)...	`OPENSHEETSET`	
Sheet list table	`SHEETLIST`	
New layout	`LAYOUTNEW`	
Page setup manager...	`PAGESETUP`	
Publish...	`PUBLISH`	
Plot...	`PLOT`	

Objects

Arch: Door S · Door L · Double · Sliding · Glass Door · 2-way Door · Fire Door · Pocket Door · Bi-fold Door · Rolling Door · Garage Door · Revolving Door · Win · Win 2 · Slide W · Fixed · Casement · Window 3-panel · Louver · Awning · Vent · Stair · Stair L · Stair U · Spiral · Half-landing Stair · Winder Stair · Ramp

Working Drawing: Auto Dim · Continue · Baseline · Rect Walls · Boundary · Door/Win Tag · Door/Win Schedule · North Arrow · Scale Bar · Elevation · Rev Cloud · MLeader · Wipeout · Annotation Scale

Room (Revit-style): Room · All rooms · Separator · By poly · Properties · Schedule · Tags · Colors · Renumber · Update

Grid / Axis: Grid / Axis · Edit grid · Grid line

Furniture: Library · Office · Home · Hotel · Hospital · Kitchen · School/Shop · Decor/Car

Piping: Gen Pipe · Clean Water · Hot Water · Waste Water · Soil Water · Vent · Tee / Join · Flip Flow

Pipe Accessories: Gate Valve · Ball Valve · Butterfly · Check Valve · PRV · Water Meter · Strainer · Backflow Prev. · Cleanout · FCO · CCO · Auto FCO/CCO · Air Vent · WHA · Pr. Gauge · Union · Reducer

Sanitary: Closet · Squat · Urinal · Washbasin · Shower · Tap · Floor Drain · Roof Drain

Pumps & Tanks: Pump · Water Tower · Ground Tank · Water Heater

Wastewater Treatment: Septic Tank · Soakaway Well

Fire Fighting: Gen Fire · Sprinkler Pipe · Hydrant Pipe / Header

Sprinkler & Hydrant: Head Pendant · Head Upright · Indoor Hydrant Box · Outdoor Hydrant Box · Hose Reel · Fire extinguisher (APAR) · Siamese · Control Box

Pump, Tank & Valve: Fire Tank · Pump Set · Electric Pump · Diesel Pump · Jockey Pump · Gate Valve · Check Valve · Alarm Check Valve · Flow Switch · Pressure Gauge

Ducting: Gen HVAC · Duct Supply Air · Duct Return Air · Duct Fresh Air · Duct Exhaust

Terminals & Equipment: 4-way Diffuser · Round Diffuser · Linear Diffuser · Return Air Grille · Exhaust Grille · Fresh Air Louver · Exhaust Fan · AHU · FCU Ducted · VAV Box · Split AC Indoor · Outdoor Unit

Dampers & Controls: Volume Damper · Fire Damper · Thermostat

Electrical installation: Gen Cable · Connect · Check Connections · Schedule · SLD · Wiring

Electrical components: kWh · MCB Box · Downlight · TL · Switch · Socket Outlet · Downlight 2 · Wall Lamp · Switch 2 · Switch 3 · Grid Switch · Stop 2 · Floor Socket · Genset

Riser & Connections: Riser · Tee / Join · Flip Flow · Floor

MEP

Menu / item	Command	Shortcut
Generate electrical installation...	`GENKABEL`	
Generate pipes...	`GENPIPA`	
Pipe specification (material)...	`PIPESPEC`	
Notasi pipa (tampil / sembunyi)...	`NOTASI`	
Simulasi animasi sistem MEP...	`SIMULASI`	
Cek bentrok jalur pipa...	`CLASH`	
Tambah legend sistem...	`LEGEND`	
Update all legends	`LEGENDUPD`	
Pasang FCO / CCO otomatis (lantai ini)	`AUTOCO`	
Generate fire fighting (sprinkler, hydrant, pump)...	`GENFIRE`	
Generate HVAC (diffusers, ducts, AHU)...	`GENHVAC`	
Panel schedule...	`JADWAL`	
Pipe isometric (automatic)	`ISO`	

Menu / item	Command	Shortcut
Pipe schematic (automatic)	`SKEMATIK`	
Pipe section (draw a cut line)	`POTONGAN`	
Isometric viewpoint…	`ISOMODE`	
Panel Single Line Diagram	`SLD`	
Panel Wiring Diagram	`WIRING`	
BOQ (Bill of Quantity)…	`BOQ`	
Derived drawings…	`VIEWS`	
Sync all derived drawings	`SINKRON`	
Delete generated results (this floor)	`HAPUSGEN`	
MEP check & summary…	`CKMEP`	
Floors (Levels)…	`LEVEL`	

Help

Menu / item	Command	Shortcut
Complete tutorial (PDF)	`TUTORIAL`	
Command list & shortcuts	`HELP`	
Contact & support…	`KONTAK`	
Community forum	`FORUM`	

24. Reference: ribbon and tool panel

Every ribbon tab with its groups and buttons, followed by the contents of the tool panel on the left.

Home

Draw: Line · Polyline · Rect · Circle · Arc · Wall · Text · Spline · Polygon · Ellipse · Point · Xline · Hatch · Donut

Block & Image: Block · Insert · Wblock · Attribute · Edit blk · Manager · Image

Modify: Move · Copy · Rotate · Mirror · Scale · Stretch · Offset · Array · Erase

Trim & Corners: Trim · Extend · Break · Join · Fillet · Chamfer · Lengthen · Explode · Match · Edit spline

Layer tools: Properties · Make current · Previous · New layer · New VP frozen · Delete layer · Merge · Purge · Off · Freeze · Lock · Isolate · Unisolate · All on · Thaw all · Unlock all

Electrical

View: VG · Electrical only · Show All

Electrical installation: Gen Cable · Connect · Check Connections · Schedule · SLD · Wiring

Electrical components: kWh · MCB Box · Downlight · TL · Switch · Socket Outlet · Downlight 2 · Wall Lamp · Switch 2 · Switch 3 · Grid Switch · Stop 2 · Floor Socket · Genset

Plumbing

View: VG · Plumbing only · Show All

Piping: Gen Pipe · Clean Water · Hot Water · Waste Water · Soil Water · Vent · Tee / Join · Flip Flow

Pipe Accessories: Gate Valve · Ball Valve · Butterfly · Check Valve · PRV · Water Meter · Strainer · Backflow Prev. · Cleanout · FCO · CCO · Auto FCO/CCO · Air Vent · WHA · Pr. Gauge · Union · Reducer

Sanitary: Closet · Squat · Urinal · Washbasin · Shower · Tap · Floor Drain · Roof Drain

Pumps & Tanks: Pump · Water Tower · Ground Tank · Water Heater

Wastewater Treatment: Septic Tank · Soakaway Well

Fire Fighting

View: VG · Fire only · Show All

Fire Fighting: Gen Fire · Sprinkler Pipe · Hydrant Pipe / Header

Sprinkler & Hydrant: Head Pendant | Head Upright | Indoor Hydrant Box | Outdoor Hydrant Box | Hose Reel | Fire extinguisher (APAR) | Siamese | Control Box

Pump, Tank & Valve: Fire Tank | Pump Set | Electric Pump | Diesel Pump | Jockey Pump | Gate Valve | Check Valve | Alarm Check Valve | Flow Switch | Pressure Gauge

HVAC

View: VG | HVAC only | Show All

Ducting: Gen HVAC | Duct Supply Air | Duct Return Air | Duct Fresh Air | Duct Exhaust

Terminals & Equipment: 4-way Diffuser | Round Diffuser | Linear Diffuser | Return Air Grille | Exhaust Grille | Fresh Air Louver | Exhaust Fan | AHU | FCU Ducted | VAV Box | Split AC Indoor | Outdoor Unit

Dampers & Controls: Volume Damper | Fire Damper | Thermostat

General MEP

Riser & Connections: Riser | Tee / Join | Flip Flow | Floor

System Drawings: Isometric | Schematic | Section | Iso Angle | 3D Orbit

Analysis & Checks: Clashes | Simulation | MEP Check | Specification | Notation | Legend

MEP documents: BOQ | Drawing | Sync | Clear Gen | IFC/OBJ

System Display: VG | Show All

View

Visibility / Graphics: VG | Plumbing only | Fire only | HVAC only | Electrical only | Show All

Elevation: 3D Orbit | Plan | Iso SW | Iso SE | Iso NE | Iso NW | Front | Right | Rear | Left | Section

Visual Styles: Visual style (VSCURRENT) | 2D Wireframe | Conceptual | Hidden | Realistic | Wireframe

Viewports: VPORTS | 1 | 2 Vertical | 2 Horizontal | 3 Right | 4 Equal | Merge

Annotation

Annotate: Aligned | Linear | Radius | Diameter | Angular | Continue | Baseline | Auto Dim | Dimstyle | Text style

Query: Distance | Area | List | ID Point | QSelect | Units

Title & Scale: Drawing title | Edit title | Text | Text styles

Hatch: Solid | ANSI31 | ANSI37 | ANSI32 | Brick | Dots | Earth | Gravel

Architecture

Arch: Door S · Door L · Double · Sliding · Glass Door · 2-way Door · Fire Door · Pocket Door · Bi-fold Door · Rolling Door · Garage Door · Revolving Door · Win · Win 2 · Slide W · Fixed · Casement · Window 3-panel · Louver · Awning · Vent · Stair · Stair L · Stair U · Spiral · Half-landing Stair · Winder Stair · Ramp

Working Drawing: Auto Dim · Continue · Baseline · Rect Walls · Boundary · Door/Win Tag · Door/Win Schedule · North Arrow · Scale Bar · Elevation · Rev Cloud · MLeader · Wipeout · Annotation Scale

Room (Revit-style): Room · All rooms · Separator · By poly · Properties · Schedule · Tags · Colors · Renumber · Update

Grid / Axis: Grid / Axis · Edit grid · Grid line

Furniture: Library · Office · Home · Hotel · Hospital · Kitchen · School/Shop · Decor/Car

Layout

Layout Management: Generator · Manager · Layout+ · Open · Model · Rename · Delete

Title Block: Title Block · Parties & Logo · Revisions · Detach

Cover & Contents: Cover · Contents · Update

Viewport & Output: Viewport · Page Setup · Plot · Publish · Sheet Set · Sheet List

Output & Manage

Output: Plot · Publish · Page · Xref · DXF · IFC/OBJ · Mview · Layout+ · Sheet set · New set · Open set · Sheet list · Purge · Aliases

Tool panel (left)

Select: Select Esc

Draw: Line L · Polyline PL · Rectangle REC · Circle C · Arc A · Ellipse EL · Polygon POL · Spline (freehand curve) SPL · Donut DO · Point PO · Construction Line XL · Hatch H

Modify: Move M · Copy CO · Rotate RO · Mirror MI · Scale SC · Stretch S · Offset O · Array AR · Erase E

Trim & Corners: Trim TR · Extend EX · Break BR · Join J · Fillet F · Chamfer CHA · Lengthen LEN · Explode X · Match Properties MA · Edit Spline SPE

Annotate: Text DT · Dimension DIM · Dimension Linear DLI · Dimension Radius DRA · Dimension Diameter DDI · Dimension Angular DAN · Drawing Title & Scale TTL

Architecture: Wall WA · Door DR · Window WI · Stair STR · Furniture FU

Room: Room (click inside walls) RM · Place All Rooms RMA · Room Separation Line RMS · Room by Polygon RMP · Room Properties RME · Room Schedule RMSCH · Show / Hide Room Tags RMT · Room Color Scheme RCS · Renumber Rooms RMN · Update Rooms to Walls RMU

Block & Image: Make Block B · Insert Block I · Write Block (file) W · Define Attributes (ATTDEF) ATT · Edit Block Attributes (ATTEDIT) ATE · Block Editor BE · Block Manager DC · Insert Image (PNG/JPG) IM · Xref Manager XREF

Sheet & Output: Sheet Set Manager SSM · New Sheet Set NSS · Sheet List Table SHEETLIST · New Layout LAYOUTNEW · Viewport (MVIEW) MV · Plot PLOT · Publish PUBLISH · Page Setup PAGESETUP

Inquiry & Utilities: Distance DI · Area AA · List LI · ID Point ID · Layer Explorer LA · Purge PU · Quick Select QSELECT · Units UN · Zoom Extents ZE · Pan P · Undo U · Redo REDO

Sanitary: Sit-down Closet CSD · Squat Closet CSJ · Urinal URN · Tap KR · Washbasin WS · Shower SH · Floor Drain FD · Roof Drain RD · Pump PM · Water Tower TRN · Ground Tank GT · Water Heater WH · Septic Tank ST · Soakaway Well SR

Electrical: Downlight T1 DL1 · Downlight T2 DL2 · TL TL · Wall Lamp WL · Switch 1 S1 · Switch 2 S2 · Switch 3 S3 · Socket 1 Gang ST1 · Socket 2 Gang ST2 · Floor Socket STF · kWh Meter KWH · MCB Box MCB · Genset GEN · Grid Switch SG · Connect (MCB-Switch-Lamp) HB

Plumbing Pipes: Clean Water Pipe PAB · Hot Water Pipe PAP · Waste Water Pipe PBE · Soil Pipe PAK · Vent Pipe PV · Reducer (insert in pipe) RDC · Tee / Join pipes TEE · Flip flow direction FLIP

Riser (All Systems): Riser (up / down floors) RS

Plumbing Accessories: Gate Valve PGV · Ball Valve PBV · Butterfly Valve PBT · Check Valve PCK · Pressure Reducing Valve (PRV) PRV · Water Meter WMT · Strainer (Y) STN · Backflow Preventer BFP · Cleanout (Clean Out) CLO · FCO (Floor Clean Out) FCO · CCO (Ceiling Clean Out) CCO · Air Vent (air release) AAV · Water Hammer Arrestor WHA · Pressure Gauge PGA · Union / Flange PUN

Fire Fighting: Sprinkler Pipe PSP · Hydrant Pipe / Header PHY · Head Sprinkler Pendant SPP · Head Sprinkler Upright SPU · Indoor Hydrant Box HYI · Outdoor Hydrant Box HYO · Hose Reel HRL · Fire extinguisher (APAR) Fire extinguisher (APAR) · Siamese Connection (FDC) FDC · Control Box (Gate Valve) VBX · Fire Tank FT · Pump Set (Diesel+Electric+Jockey) PMS · Electric Pump PME · Diesel Pump PMD · Jockey Pump PMJ · Gate Valve GV · Check Valve CV · Alarm Check Valve ACV · Flow Switch FS · Pressure Gauge PG

HVAC: Duct Supply Air DSA · Duct Return Air DRT · Duct Fresh Air DFA · Duct Exhaust DEA · 4-way Supply Diffuser DFS · Round Supply Diffuser DFR · Linear Supply Diffuser DFL · Return Air Grille RG · Exhaust Grille EG · Fresh Air Louver LOU · Exhaust Fan EF · AHU AHU · FCU Ducted FCU

VAV Box VAV | Split AC Indoor SAC | Outdoor Unit ODU | Volume Damper VCD | Fire Damper FDM | Thermostat TH

File: Sample plan SA